13th Bangladesh Physics Olympiad 2023

National Round Exam

Category-D

Understanding White Dwarfs

White dwarfs are the end state of stars that fail to collapse into a black hole (this is a very tough statement, we will make it precise below). For white dwarfs, the gravitational inward pressure is countered by the relativistic degeneracy pressure of a Fermi gas, which, in the case of white dwarfs, is provided by the electrons in the matter. We first derive the equation of state for such a gas.

The relation of energy E and momentum p for a relativistic particle is given by:

$$E = pc = \hbar kc$$

where we have used the de Broglie relation in the second step. Needless to state, p is the magnitude of the 3-dimensional momentum p of the particle. Recall that the spin of the electron is $1/2$ and thus the level of degeneracy (the number of electrons in the same energy state) is 2.

Now, the phase space is covered by the coordinates and momentum of any particle. So, for instance, in a 2-dimensional coordinate space, one also has two directions of momentum, and therefore the phase space is 4-dimensional: two coordinates and two momenta, to reiterate. In quantum statistics, it is given that the "volume" of phase space associated with each direction is given by $\hbar$. Thus, for the four-dimensional phase space, this "volume" happens to be $\hbar^2$.

Since electrons are fermions, they satisfy the Pauli exclusion principle, which requires each particle to have different "tags." So, even if two quantum particles can have the same position, they must have different momentum. Therefore, fermions at their ground (minimum energy) state have to have their momentum distributed inside a constant energy surface, which we call the Fermi surface. For example, in two dimensions and for non-relativistic free particles, this surface is given by:

$$E_F = \frac{\mathbf{p_F} \cdot \mathbf{p_F}}{2m}$$

where $\mathbf{p_F}$ is the Fermi momentum and E_F is the Fermi energy. The total number of particles that can be put inside the Fermi surface is found in a straightforward manner by utilizing all the information given above.

(a) Find the total number of electrons inside the Fermi surface given by the Fermi momentum p_F for the electrons. **[3]**

(b) Find the total energy of the system described by the configuration in part (a). Find the relationship between the energy density of the system and the number density of the system. Remember that the number density is the number of particles (electrons in this case) per unit volume. **[4]**

(c) Using your knowledge of Thermodynamics, find the pressure of the relativistic gas of fermions in terms of the density of the gas. **[2]**

Gravitational Pressure inside stars

Next, we derive the inward pressure inside the star caused by its mass distribution. We can determine it following the steps described below.

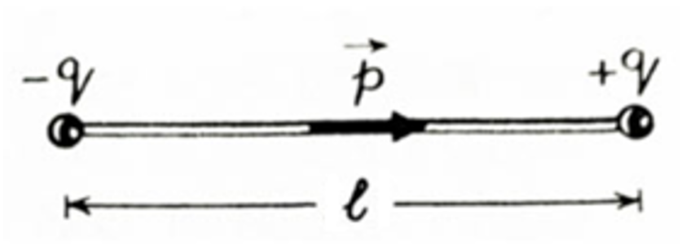
(a) Consider a spherical ball of radius R which has a spherically symmetric mass density $\rho(r)$, where r is the distance measured from the center of the ball. Now, a point mass m is held inside the ball at a distance $r \leq R$. Find the expression of the force on this point mass due to the spherical ball at that distance. **[2]**

(b) Now, think of two concentric spheres (both inside the ball) with radii r and $r + dr$ with the pressure on them P and $P + dP$ on them respectively. Find the net (outward) force on the shell of matter in terms of the pressure difference dP and related quantities. **[3]**

(c) Using the condition that the net outward force is compensating the inward force due to gravity, find the differential equation that governs a star created out of relativistic electron gas, discussed in problem 1, where one found the relation between the pressure and the density of the relativistic electron at a point. **[3]**

Chemistry of Dipoles

When a positive and negative charge of equal magnitude are separated by a very small fixed distance, the setup is called a dipole.



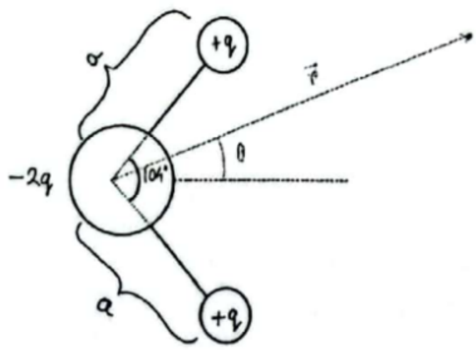
In order to conveniently describe the behavior of the dipole, we will define a quantity $\vec{p} = q\ell\hat{\ell}$. The vector $\vec{p}$ points from the negative charge $-Q$ to the positive charge $+Q$. This is useful because the electric potential at a point $\vec{r}$ very far away from the dipole (measured from the center of the dipole, i.e. the midpoint between the charges) is given by

$$V = \frac{k\vec{p} \cdot \vec{r}}{r^3}$$

where k is Coulomb's constant and $\vec{r}$ is the position vector of the point. The dot product between the two vectors is defined as $\vec{p} \cdot \vec{r} = pr\cos(\theta)$, where θ is the angle between the two vectors $\vec{p}$ and $\vec{r}$.

Part A : Potential due to water molecules

For this problem, we will model the $\mathrm{H_2O}$ molecule as a combination of one $-2q$ and two $+q$ charges, as shown in the figure below.

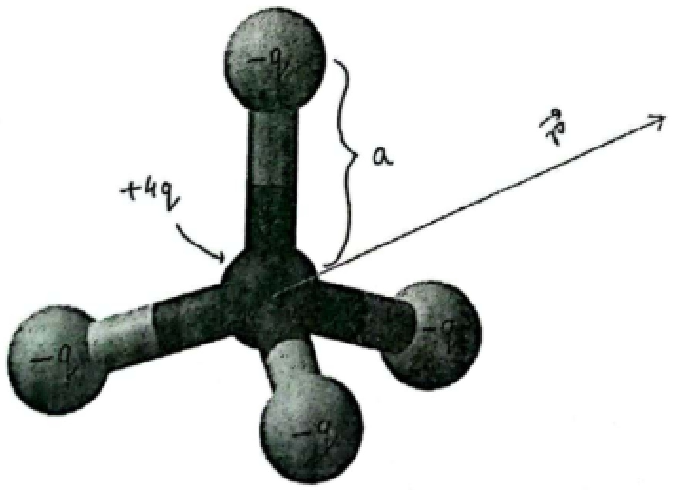


(A$_1$) Consider a point P at a distance $\vec{r}$ relative to the oxygen atom that makes an angle θ with the axis of the water atom, as shown in the figure above. Find the distances between this point P and midpoint of each hydrogen-oxygen bond, in terms of r and θ. **[1]**

(A$_2$) Find the electric potential at that point P assuming $r \gg a$, in terms of r, q, and θ. **[2]**

Part B: Potential due to a CCl_4 molecule

A carbon tetrachloride molecule, on the other hand, has a symmetric tetrahedral shape, as shown in the figure below. We can model this atom as $+4q$ charge at the center and $-q$ charge at each of the four corners.



(B) Once again, find the potential V', far away from the atom $r \gg a$. **[2]**

Part C: The H_2O revisited

Let's go back to our H_2O model from part A. This time, we want to calculate the electric field due to a single water molecule.

(C) Find the electric field $\vec{E}(r, \theta)$ due to this water atom at $r \gg a$. **[2]**

Hint: The radical and tangential components of the electric field are given by

$$E_r = -\frac{\partial V}{\partial r}$$
$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta}$$

where V is the electric potential, r is the position vector, and $\frac{\partial V}{\partial r}$ means the partial derivative of V with respect to r (in simple words, A derivative with respect to r, where you consider other variables except r to be constant.)